

Hamud Abdul Aziz

South Jakarta, Jakarta, Indonesia

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SUMMARY

An undergraduate Computer Science student with a strong interest in data analysis and intelligent systems. Experienced in working with image processing concepts and building problem-solving models. Eager to leverage analytical skills and a detail-oriented mindset to help derive actionable insights from data in a professional team.

EDUCATION

2023 - Present	Bachelor of Computer Science at Universitas Padjadjaran	(GPA: 3.80/4.0)
2020 - 2023	Multimedia at SMKN 59 Jakarta	(Grades: 90.98)

EXPERIENCES

Department of Computer Science Unpad	Feb 2025 – June 2026
<i>Laboratory Teaching Assistant</i>	

- Supported laboratory instruction across three undergraduate courses: Database Systems (MySQL, ER modeling, relational design), Algorithms & Programming (algorithm design, C++ fundamentals), Object-Oriented Programming (Java, C++, Python), and Computer Networking(Cisco, Networking).
- Mentored and evaluated 100+ students, contributing to an 85% course passing rate through structured lab guidance and technical support.
- Developed quizzes, debugging exercises, and supervised assessments while collaborating with lecturers to standardize grading rubrics across lab sections.

Center for Innovation in Teaching and Learning (PIPP) Unpad	Oct 2025 - Jan 2026
<i>Educational Simulation Developer (Intern)</i>	

- Contributed to the development of a full-scale 3D campus simulation environment representing Universitas Padjadjaran for educational and visualization purposes.
- Designed and built detailed 3D architectural assets and building models, ensuring spatial accuracy and visual consistency.
- Collaborated within a 9-member cross-functional team to implement environment assets and optimize scene performance.
- Supported environment integration, layout structuring, and asset refinement to enhance user immersion and navigational clarity.

RDAT, BEM Kema FMIPA Unpad	May 2025 – Jan 2026
<i>Backend Lead & Co-Lead Consultant Research Team</i>	

- Served as Backend Lead for the BEM FMIPA website, responsible for API development, database design, and system architecture decisions.
- Led backend feature implementation and coordinated with frontend and design teams to ensure scalable and maintainable deployment.
- Appointed as Co-Vice Consultant for Research, supporting research project development, methodology structuring, and analytical review within the bureau.
- Contributed to strategic discussions on technology integration and data-driven decision-making for organizational programs.

SELECTED PROJECTS

NutriMatch: AI-Powered Personalized Diet & Allergen-Smart Recommendation System Apr 2026 – Jun 2026

AI Engineer, Coding Camp 2026 powered by DBS Foundation

- Analyzed and processed multi-dimensional user health data (age, gender, BMI, activity levels, and diet goals) to build automated calculators for BMR, TDEE, and daily caloric targets.
- Designed and evaluated a multi-class Deep Learning classification model using TensorFlow/Keras to cross-reference dense food ingredient datasets against user allergy profiles.
- Mitigated critical medical risks by driving model optimization based on the Recall metric, successfully minimizing False Negatives to ensure hazardous allergens were strictly filtered out
- Developed data-driven insight engines by engineering custom training pipelines and wrapping the predictive models into live endpoints, enabling automated, personalized 7-day meal plan recommendations.

BeliBiji - E-Commerce & Biodiversity Crowdfunding Platform Mar 2026 – Jun 2026

Backend Developer

- Designed and developed a secure, scalable RESTful API architecture for an e-commerce and ecological crowdfunding web application using Node.js/Express and PostgreSQL.
- Built robust backend core services for inventory control and supply management, facilitating seamless stock submission flows and validation for local farmers.
- Integrated third-party Payment Gateway APIs in Sandbox mode to efficiently process automatic B2C shopping checkouts and micro-donation transactions.
- Developed an AI-driven product review analysis system using a Decision Tree model and built its backend endpoint to process textual feedback parameters, enabling automated sentiment and rating classification.
- Implemented the system iteratively under the Agile/Scrum framework, performing comprehensive unit and integration testing to ensure API reliability.

Enterprise Network Infrastructure for Film Production Studio May 2026 – Jun 2026

Network Architect & Designer

- Designed high-speed end-to-end network topologies to support massive 4K/8K video data transmissions and secure cross-city collaboration between Jakarta and Bandung studios.
- Implemented layered security and traffic isolation by segmenting the LAN into 4 distinct VLANs using a Layer 3 Core Switch and enforcing strict data-access restrictions via Extended ACLs.
- Established secure inter-branch WAN connections by configuring IPsec VPN over GRE Tunnels with AES and SHA-1 encryption over a simulated public ISP infrastructure.
- Mitigated critical bandwidth bottlenecks by migrating the legacy WAN interface to full GigabitEthernet architecture, successfully achieving 0% packet loss and optimizing real-time file transfer stability.

KotakP3K (MediProfen): Virtual Reality Emergency Medical Training Simulation Mar 2026 – Jun 2026

3D Animator & UI Designer

- Conceptualized and developed 8 distinct medical-response animations using Blender to realistically simulate real-time clinical patient symptoms and trauma reactions.
- Designed lifelike character rigging and visual asset feedback loops for critical scenarios, including choking indicators, respiratory distress reactions, and post-injection recoveries.
- Engineered and cross-referenced visual animation keys to synchronize seamlessly with Unity's XR Interaction Toolkit triggers based on real-time user controller inputs.
- Collaborated within a 5-member cross-functional team to integrate 3D environmental assets, logic-driven scenario state machines, and gamified user evaluation systems.

SKILLS

Languages	English(Fluent), Bahasa Indonesia(Native)
Programming Languages	C++, C#, Python, Javascript, PHP (CodeIgniter, Laravel)
Tools and Libraries	VSCode, Git, Unity, MySQL, Figma.